

# Fast and Accurate Decomposition of Deterministic Jitter Components in High-Speed Links

Yan Duan<sup>ID</sup>, *Student Member, IEEE* and Degang Chen<sup>ID</sup>, *Fellow, IEEE*

**Abstract**—In a high-speed digital communication system, jitter performance plays a crucial role in bit-error rate. It is important to accurately derive each type of jitter as well as total jitter and to identify root causes of jitter by jitter decomposition as a key tool. In this paper, we extend our previously reported algorithm to decompose duty cycle distortion as well as other deterministic jitter components including periodic jitter (PJ), intersymbol interference (ISI), and random jitter (RJ), and describe more detail of this algorithm with extensively simulation and measurements to verify the algorithm. The proposed ISI model only requires the time invariant condition, which is true for almost all systems, while the traditional ISI model is valid only with linear time invariant assumption. Compared to the conventional methods, the proposed jitter decomposition method is able to obtain estimated individual jitter component values with good accuracy by using fewer samples of total jitter data. The efficiency and accuracy of the proposed method were demonstrated by simulation and hardware experiments.

**Index Terms**—Comparator network, high-speed data links, jitter decomposition, time interval error (TIE).

## I. INTRODUCTION

THE data processing capabilities of integrated circuits increased greatly due to scaling of integrated circuit technology in the past decade [1]. However, digital data communication circuit has become one of the dominant reasons to cause bottlenecks in these systems. To overcome these bottlenecks, more sophisticated high-speed link circuits have replaced those simple input/output (I/O) drivers integrated in these chips. The aggressive scaling of I/O bandwidth demands double every two to three years on average [1]. However, the I/O performance is limited by electrical channel bandwidth limits. Meanwhile, as the data rate increases, the intersymbol interference (ISI) becomes severe due to the bandwidth limitation. Jitter and noise, generated inevitably in the transmitter, channel, and receiver, impact the performance of the system. Accurate jitter analysis is crucial for next-generation I/O design in order to obtain an acceptable bit-error rate (BER).

Jitter is defined as the deviation of transition edges from their ideal edge to the actual one in time domain. An observed total

jitter (TJ) distribution includes different components [2], [3]. It can basically be decomposed into a bounded deterministic jitter (DJ) and an unbounded random jitter (RJ). RJ is usually considered as Gaussian, unbounded probability behavior. It is observed at both distribution tails, extending them toward infinity. DJ could be of arbitrary shape and be expressed by various subcomponents in order to distinguish various root causes. DJ is further divided into PJ, bounded uncorrelated (BUJ), and data-dependent jitter (DDJ). DDJ is related to transmitted data patterns. DDJ consists of duty cycle distortion (DCD) and ISI. DCD is caused by difference in the pulse width between high and low logical levels which are caused by voltage offsets or different rise and fall times during signal transitions. BUJ is always considered bounded because of the limited coupling strength.

Understanding the amount of jitter generated by each jitter source is imperative for predicting overall system performance [1]. A great deal of researchers have proposed different approaches to decompose jitter in the past decades. In [4], Li proposed a tail fitting algorithm based on the Gaussian tail model by using probability distribution of collected jitter values. In this model, RJ could be observed at the outer tails of a TJ distribution. In [5]–[8], researchers developed different approaches to decompose RJ and DJ components with different tail fitting algorithms. However, these tail fitting methods requires a large amount of jitter samples to correctly identify the fit tail part of distribution. In [9], the proposed method relies on an idea that a TJ probability density function (PDF) is given as convolution result of the DJ and RJ components. If one component can be approximately estimated first, other components can be estimated by the deconvolution algorithm and thus to retrieve the Gaussian model parameters. However, this method lacks accuracy since either DJ or RJ component has to be roughly estimated prior to the deconvolution.

One category of decomposition methods is through frequency analysis. Based on the time domain, series of jitter can be represented in the frequency domain by the Fourier transform (FT), many researchers proposed frequency domain methods [10]–[13]. Among these methods, power spectral density (PSD) represents jitter spectrum and peaks in the spectrum can be interpreted as PJ or DDJ, while the average noise floor is the power of RJ.

Another jitter analysis technique is based on time-domain analysis [14]–[16] which relies on jitter measurements carried out in a real-time mode and is used in those dedicated real-time measurement systems, such as high-speed sampling scopes or time interval analyzer (TIA). In [17], Xu proposed a low-cost

Manuscript received October 16, 2017; revised December 12, 2017 and January 7, 2018; accepted January 17, 2018. Date of publication February 6, 2018; date of current version November 15, 2018. (Corresponding author: Yan Duan.)

Y. Duan is with the Department of Electrical and Computer Engineering, Iowa State University, Ames, IA 50011 USA (yanduan@iastate.edu).

D. Chen is with the Iowa State University, Ames, IA 50011 USA (djchen@iastate.edu).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TEM.2018.2797122

jitter separation method based on analog-to-digital convertor testing. However, this method is limited to separate jitter in only clock channels with low speeds. All the methods mentioned above need large data samples, for which sufficient memory depth is needed and hence the accuracy can be guaranteed through these digital signal processing (DSP) techniques [18].

Among all jitter components, ISI is an important jitter component needed to be addressed. ISI is the mainly contributor to TJ and deep investigations are needed. Some researchers have proposed the ISI modeling in the past. One popular idea of modeling the channel is as a first-order and a second-order low-pass filter [20], [21]. However, such a model is too simple to represent the real channel and becomes invalid in presence of discontinuities. Another widely used ISI modeling method is based on the convolution technique [16]. However, it is very time-consuming since the cursor is usually 100-bit long.

In order to solve the jitter decomposition more efficiently and with fewer samples, a jitter decomposition algorithm using least squares (LS) was previously proposed by our group [19]. This algorithm simultaneously separates ISI, RJ, and PJ and includes a new time-domain ISI model based on channel pulse responses. The ISI model is more effective than these conventional cursor convolution techniques. We extended the algorithm to decompose DCD as well as other deterministic jitter components in [19] and describe more detail of this algorithm with extensively simulation and measurement to verify the algorithm. It has several advantages. First, it can accurately estimate PJ, ISI, DCD, and RJ. Second, it needs fewer data samples than the instrument test which saves the memory requirement. Third, the new ISI modeling is accurate and simple to ISI jitter estimation and provide ISI jitter information in detail. One of the main contributions of this algorithm is the novel ISI modeling. It is based on channel pulse response in time domain, which is simpler than the conventional ISI cursors convolution technique. The proposed ISI modeling is more accurate and realistic than the low-pass filter model. The LS was used to obtain parameter values in the above-mentioned jitter model [22], [23]. A great deal of simulation and measurements show that the proposed method with great accuracy in the jitter decomposition when compared with the simulation and conventional instrument test methods.

The rest of this paper is organized as follows. In Section II, the conventional PJ, DCD, and RJ are reviewed; the new ISI model is addressed. In Section III, the total jitter model is described. An LS method is applied to total jitter model and the estimation of jitter component is derived. In Section IV, the validation for the ISI model and jitter decomposition simulation result are presented. In Section V, measurement results will show the validation of our proposed methods and Section VI will present the conclusions of the paper.

## II. JITTER MODELING

In this section, we present the PJ, DCD, ISI, and RJ models in detail. PJ, DCD, and RJ models are widely used in jitter analysis. A new time domain ISI jitter model for a lossy channel is developed for efficient and accurate ISI estimation.

### A. Periodic Jitter, Duty Cycle Distortion and Random Jitter Modeling

PJ, DCD, and RJ are modeled by the traditional method [1], [16]. PJ is usually the result of a cross coupling or electrical migration interfere (EMI) problem in the system. PJ is a repeating jitter from a switching power supply coupling into the data or phase-locked loop (PLL) reference clock feed through and its frequency typically known from spectral analysis. It is modeled mathematically as a sum of sinusoids. Here, a single sinusoid (fundamental frequency) is used as an example given by the following equation:

$$\begin{aligned}\Delta t_{PJ}[n] &= A \sin(2\pi f_0(t - nT) + \phi) \\ &= a \sin\left(\frac{2\pi f_0 n}{f_s}\right) + b \cos\left(\frac{2\pi f_0 n}{f_s}\right)\end{aligned}\quad (1)$$

where  $\Delta t_{PJ}[n]$  is a PJ amount at sampling time  $nT$ ;  $f_0$  represents the fundamental frequency of PJ;  $A$  is the amplitude of PJ; In a real system, PJ can be power supply induced noise or input reference clock of PLL and can be obtained by spectral analysis as the state of the art [1]. In this paper, we considered the PJ from reference clock of PLL as an example in simulation;  $f_s$  is the frequency of data stream, and  $\phi$  is the initial phase of PJ.  $a$  and  $b$  are the estimation PJ parameters.

We also extend the algorithm to decompose DCD as well as other deterministic jitter components in [19]. There are two primary causes of DCD: one is the transmitter threshold offset from its ideal level, which causes the output of the transmitter with the DCD; the other cause of DCD is asymmetry in rising and falling edge of clock path.

DCD causes the widest set of frequency components in the clock pattern and is viewed as a series of adjacent positive and negative impulses at the input. Using the jitter spectrum display, the DCD component of jitter is half of the data rate, which can be modeled by the following equation [16]:

$$\begin{aligned}\Delta t_{DCD}[n] &= J_{DCD} \times \cos(n\pi) \\ &= [-J_{DCD}, J_{DCD}, -J_{DCD}, J_{DCD}, \dots]\end{aligned}\quad (2)$$

where  $\Delta t_{DCD}[n]$  is a DCD at sampling time  $nT$ ;  $J_{DCD}$  is the DCD amplitude which is to be estimated in the algorithm.

RJ is created by unbounded jitter sources, such as Gaussian white noise. Thermal noise, shot noise, and other high-order noise processes are the typical sources to cause this type of jitter. The statistical PDF for RJ is generally treated as a Gaussian distribution. The PDF for random jitter is given by

$$f_{RJ}(\Delta t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\Delta t - \mu)^2}{2\sigma^2}\right)\quad (3)$$

where  $\mu$  represents the mean and  $\sigma$  is the standard deviation of Gaussian distribution. In the algorithm, the  $\mu$  is zero and  $\sigma$  is the estimation RJ parameter.

### B. New ISI Jitter Modeling

ISI is usually the result of a bandwidth limitation problem of channel, which has a low-pass filter characteristic. Due to a reduction in channel bandwidth, limited rise and fall times of

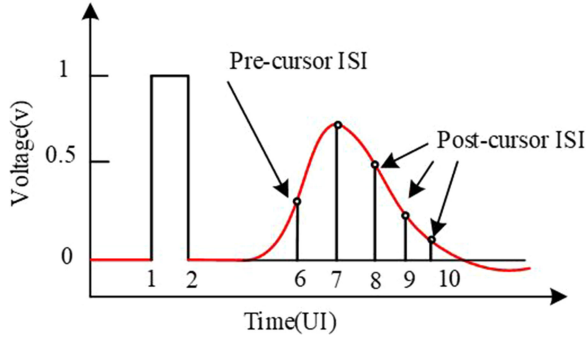


Fig. 1. Pulse response of the channel.

the signal will cause varying amplitudes of data bits, these data bits depend on repeating-bit lengths and preceding bit states.

Traditionally, the ISI PDF model is based on a commonly used convolution technique. The PDF calculation is based on a single-bit response (SBR). Fig. 1 shows an SBR after a lossy channel. The input nice narrow pulse becomes widened and attenuated at the output of the channel and it occupies the precursor and postcursor samples. The dots of the received pulse represent the symbol-spaced samples. However, this process is time consuming when the number of considered cursor is large. For example, typical single-bit responses are often more than 100-bit long [16]. When the nonlinearity of system is severe or the data pattern is nonwhite, this convolution approach may fail. Due to the limitation of the ISI cursor modeling for ISI decomposition, we developed a time domain ISI jitter model.

Since some kinds of equalizers are used to properly equalize to the precursor in a high-speed link, the post cursors from the previous  $k$ -bit is considered to model the ISI effect.

The postcursor  $k$  depends on the data rate and the amount of loss of the channel, which can be determined during channel characterization. An example is used to illustrate the time-domain ISI model. We use the data sequence  $b_1 \dots b_5$ , as an example, shown in Fig. 2(a) and (b). The black curve is the ideal data sequence and blue curve is the actual data sequence after a lossy channel. Assume that the postcursor number in this channel is 4,  $b_5$  is the current bit, and  $b_1 b_2 b_3 b_4$  are preceding 4 b (4-b postcursor).  $b_1 b_2 b_3 b_4$  has 16 binary pattern combinations. If the combination of  $b_1 b_2 b_3 b_4$  is 0101, as shown in Fig. 2(a), the time deviation of actual  $b_5$  edge and the ideal  $b_5$  edge is defined as an ISI induced jitter  $J_5$ . The index 5 is the decimal representation of 0101. If the combination of  $b_1 b_2 b_3 b_4$  is 1011, as shown in Fig. 2(b), the time deviation of actual  $b_5$  edge and the ideal  $b_5$  edge is noted as jitter  $J_{11}$ . The index 11 is the decimal representation of 1011. Different  $b_1 b_2 b_3 b_4$  binary combinations generate different ISI jitter amount to current bit  $b_1$ . A long data sequence  $b_{-3} b_{-2} \dots b_{M-1} b_M$  shown in Fig. 2(c) was used to show the ISI model in detail. The actual data  $b_{-3} b_{-2} b_{-1} b_0$  0101 is considered as block 1, which generates  $J_5$  to bit  $b_1$ . The actual data  $b_{-2} b_{-1} b_0 b_1$  1011 is considered as block 2 which causes  $J_{11}$  to bit  $b_2$  and the data  $b_{-1} b_0 b_1 b_2$  0110 is considered as block 3 which generates  $J_6$  to bit  $b_3$ . We divided all the data bits into different blocks and each block generate different jitter to the next bit. Table I describes the ISI model parameters and includes

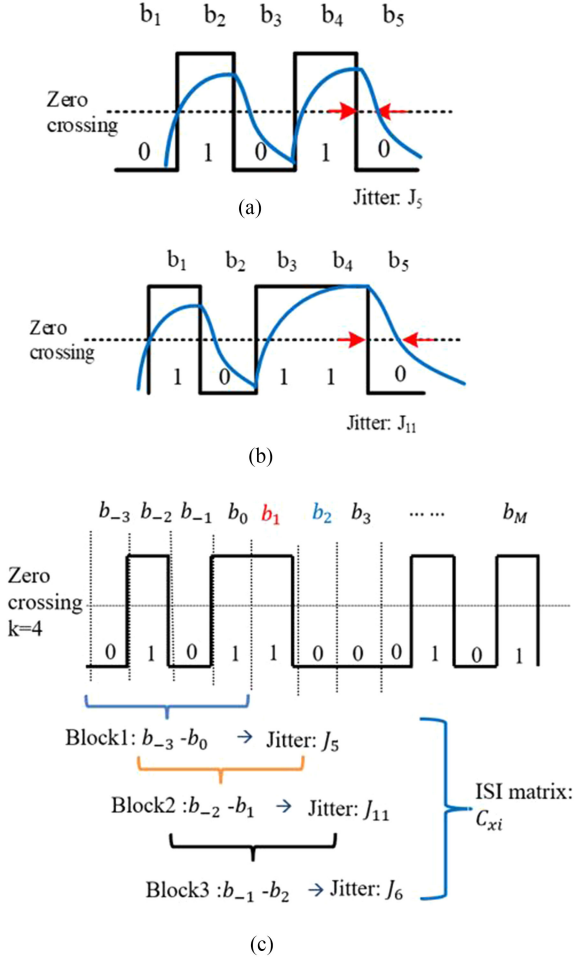


Fig. 2. ISI jitter modeling (a), (b), and (c).

TABLE I  
PARAMETERS OF ISI MODEL

| $b_1$ - $b_4$ binary combinations | Corresponding Jitter | Corresponding sign |
|-----------------------------------|----------------------|--------------------|
| 0000                              | $J_0$                | $C_0$              |
| ...                               | ...                  | ...                |
| 1011                              | $J_{11}$             | $C_{11}$           |
| ...                               | ...                  | ...                |
| 1111                              | $J_{15}$             | $C_{15}$           |

all binary combinations, corresponding jitter  $J$ , and a sign  $C$ . In this example, the 4-b post cursor has 16 binary combinations and every binary combination generates a corresponding jitter value  $J$ . The corresponding sign  $C$  is used to determine which kind of binary combinations of 4-bit post cursor for the current bit. In Fig. 2(a), the 4-bit binary combination is 0101 and the corresponding jitter value is  $J_5$ , the corresponding sign  $C_5$  is equal to 1 while other corresponding signs are all zeros.

A formula is used to describe the ISI jitter model based on the observation that different postcursor binary combinations generate different ISI jitter on the current bit. The formula is

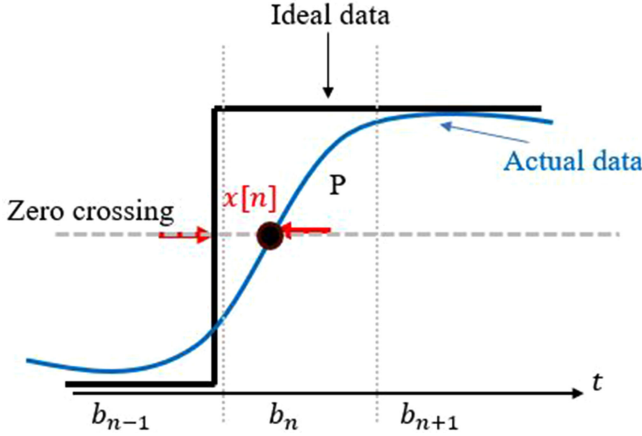


Fig. 3. Definition of TIE.

given by

$$\Delta t_{\text{ISI}}[n] = \sum_{l=0}^{2^k-1} (J_l \times C_l[n]) \quad (4)$$

$$C_l = \begin{cases} 1, & \text{if binary to decimal } b_{(n-k)}b_{(n-k+1)} \dots b_{(n-1)} = l \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where  $\Delta t_{\text{ISI}}[n]$  denoted as an ISI jitter of data bit  $n$  at sample time  $nT$ .  $b_n$  is the current bit and its jitter is determined by binary combinations of previous  $k$  bits from  $b_{n-k}$  to  $b_{n-1}$ .  $l$  means the decimal number of binary combination  $b_{n-k}b_{n-k+1} \dots b_{n-1}$ .  $J_l$  represents the jitter value of the  $l$ th binary combination for the current bit  $b_n$ .  $C_l$  is a corresponding sign to represent the binary combination of previous  $k$ -bit. Equation (4) shows that ISI jitter of the current  $b_n$  is the jitter amount of the previous  $k$ -bit binary combination induced jitter. This model is not dependent on any linearity or superposition requirements of a system and the ISI jitter is only dependent on the previous  $k$ -bit binary combination, which makes the ISI model more robust to channel nonidealities.  $J_l$  is the ISI estimation parameter in the algorithm.

### III. JITTER DECOMPOSITION ALGORITHM

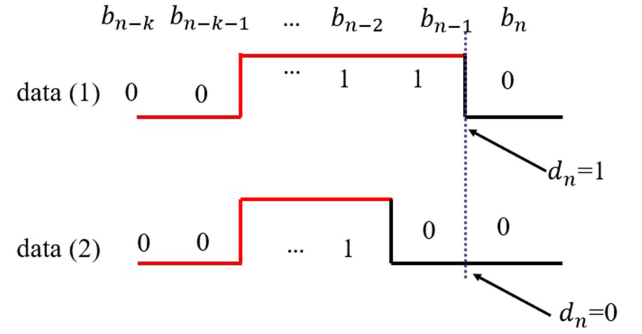
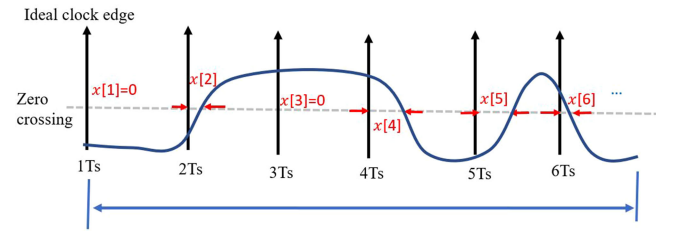
In this section, we describe the time interval error (TIE) modeling used in this paper, and then develop the details of the decomposition algorithm.

#### A. Time Interval Error

For a linear time invariant (LTI) system, the total jitter in time domain is considered as a linear sum of DJ components and the square root of the RJ components [1]. All DJ components, as well as RJ ones, contribute to total jitter [1], which results in TIE. TIE is defined as the timing difference between the zero-crossing time of the ideal data and that of the actual data, which is denoted by  $x[n]$  for bit  $b_n$  in Fig. 3.

With the proposed ISI jitter modeling, the TIE in bit  $n$  with PJ, DCD, RJ, and ISI can be simulated as

$$x[n] = d_n \times [\Delta t_{\text{PJ}}[n] + \Delta t_{\text{DCD}}[n] + \Delta t_{\text{ISI}}[n] + \Delta t_{\text{RJ}}[n]] \quad (6)$$

Fig. 4. Data transition sign  $d_n$ .Fig. 5.  $M$  bit TIE sequence.

where  $x[n]$  means the TIE amount at sampling time  $nT$ . Parameters  $a, b, f_s, f_0, C_l, J_l$  and  $J_{\text{DCD}}$  have been described in previous section,  $\delta[n]$  is pure RJ in sampling time instance  $nT$ .  $d_n$  is data transition sign which indicates the existence of or transition from bit  $n-1$  to bit  $n$ .  $d_n$  is 1 when the falling or rising edge occurs from bit  $n-1$  to bit  $n$ , as shown in data (1) of Fig. 4.  $d_n$  is 0 when no data transition occurs from bit  $n-1$  to bit  $n$ , as shown in data (2) of Fig. 4. In this case, no jitter exists for the current bit  $n$ .

#### B. Jitter Decomposition by LS

We used an LS-based decomposition method to estimate the parameters in the proposed model. It is known that TIE jitter is a linear equation from (6). For a linear time invariant system, the LS estimation overcomes the convergence problem [22], [23] and does not require any special distribution property for inputs. Because of this characteristic, we applied the LS to estimate the PJ, DCD, RJ, and ISI parameters  $[abJ_{\text{DCD}}J_0J_1 \dots J_{(2^k-1)}]$ .

In order to obtain the estimation of PJ, DCD, ISI, and RJ, we define that the length of a TIE sequence is  $M$ , as shown in Fig. 5. The TJ in each bit is  $x[1], x[2], \dots, x[M]$  taken at sampling time  $1T, 2T, \dots, MT$ , respectively. Then the total jitter sequence  $Z_M$  can be expressed by the following equation [22]:

$$Z_M = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(M) \end{bmatrix} = H_M \theta + V_M \quad (7)$$

where  $V_M$  represents the RJ vector (denoted as  $[\delta[1] \delta[2] \dots \delta[M]]^T$ ) for the jitter sequence,  $\theta = [a \ b \ J_{\text{DCD}} \ J_0 \ J_1 \dots J_{(2^k-1)}]^T$  which is the estimation parameters for PJ, DCD, and ISI.  $H_M$  is the coefficient matrix for the whole jitter



sequence. The submatrix  $A$  in  $H_M$  is PJ coefficients matrix. The submatrix  $B$  is DCD coefficients and submatrix  $C$  is ISI coefficients matrix

$$H_M = [A \ B \ C]$$

$$A = \begin{bmatrix} d_1 \times \sin\left(\frac{2\pi f_0}{f_s}\right) & d_1 \times \cos\left(\frac{2\pi f_0}{f_s}\right) & \cdots & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ d_M \times \sin\left(\frac{2\pi f_0 M}{f_s}\right) & d_M \times \cos\left(\frac{2\pi f_0 M}{f_s}\right) & \cdots & \cdots \end{bmatrix}$$

$$B = \begin{bmatrix} d_1 \times \cos(\pi) \\ d_2 \times \cos(\pi) \\ \vdots \\ d_M \times \cos(\pi) \end{bmatrix}$$

$$C = \begin{bmatrix} d_1 \times C_{01} & \cdots & d_1 \times C_{2^k-11} \\ d_2 \times C_{02} & \cdots & d_1 \times C_{2^k-12} \\ \vdots & \ddots & \vdots \\ d_M \times C_{0M} & \cdots & d_M \times C_{2^k-1M} \end{bmatrix}. \quad (8)$$

In (8), the coefficient  $C_{xi}$  of submatrix  $C$  is determined by data pattern, which can be obtained from the data stream.  $i$  is the  $i$ th data bit and  $x$  is the decimal number of binary combination of previous  $k$  bits of  $i$ th data bit. For example, let us assume a channel which the post cursor number  $k$  is 4.  $C_{2i}$  means the corresponding sign of the  $i$ th bit data when the binary combination of previous 4 b is 0010. In order to obtain submatrix  $C$ , the data sequence ( $M + 4$  b length)  $b_{-3}b_{-2} \dots b_{M-1}b_M$ , as shown in Fig. 2(c), was divided into  $M$  blocks and each block is of 4 b. In block 1, the data  $b_{-3}b_{-2}b_{-1}b_0$  0101 generates ISI jitter  $J_5$  to bit  $b_1$ . Then the  $C_{51}$  is 1 while other  $C_{xi}$ s are 0. In block 2, the data  $b_{-2}b_{-1}b_0b_1$  1011 generates ISI jitter  $J_7$  to bit  $b_2$ . Thus, the  $C_{72}$  is 1 while other  $C_{xi}$ s are 0. For the PJ, we assume that PJ mainly comes from PLL in our analysis or perform FFT to jitter sequence. In the spectrum, the frequency of maximum amount of the jitter is PJ frequency  $f_0$ .  $f_s$  is known parameters from the previously mentioned section. The data length  $M$  is determined by the PJ frequency  $f_0$ , data rate  $f_s$ , and data pattern pseudo-random binary sequence (PRBS-n). It should include integer periods ( $w$ ) of PJ at least one period and integer periods ( $y$ ) of PRBS-n full data pattern at least one period which is given by the following equation:

$$w = M \times \frac{f_0}{f_s}$$

$$y = \frac{M}{2^n} \quad (9)$$

For example, when the PJ  $f_0$  is 100 MHz, data rate  $f_s$  is 10 Gb/s and the data pattern is PRBS-7, the  $M$  should be at least 127 b. We use 1.27 kb in the simulation.

The solution  $\theta_M$  for length of jitter sequence  $M$  is

$$\hat{\theta} = [H_M^T H_M]^{-1} H_M^T Z_M. \quad (10)$$

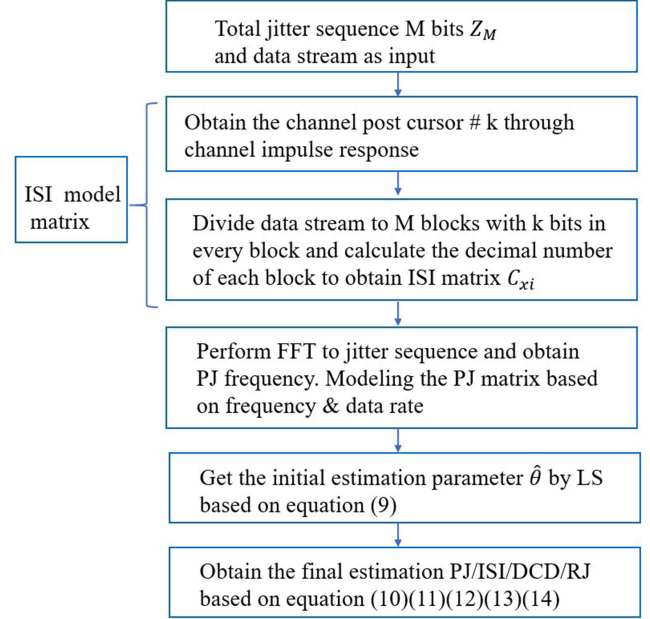


Fig. 6. Flow chart of the proposed algorithm.

The estimation parameter  $\hat{\theta}$  is

$$\hat{\theta} = [\hat{a}, \hat{b}, \hat{J}_{\text{DCD}}, \hat{J}_0, \hat{J}_1, \dots, \hat{J}_{2^k-1}]. \quad (11)$$

With the estimation results, the amplitude of the PJ can be calculated as

$$\hat{A} = \sqrt{\hat{a}^2 + \hat{b}^2}. \quad (12)$$

The ISI is calculated by

$$\widehat{\text{ISI}}_{\text{pk-pk}} = \max(\hat{J}_0, \hat{J}_1, \dots, \hat{J}_{2^k-1}) - \min(\hat{J}_0, \hat{J}_1, \dots, \hat{J}_{2^k-1}). \quad (13)$$

The variance of RJ is

$$\sigma^2 = \text{var}(Z_M - H_M \hat{\theta}). \quad (14)$$

The flow chart of the proposed algorithm is given in Fig. 6. The TIE sequence  $M$  bits with PJ, ISI, DCD, and RJ and data stream which generated the corresponding  $M$  bits TIE sequence are the input of the algorithm. First, we find each submatrix in (8). We performed the inverse Fast Fourier transform (IFFT) on the  $s$ -parameter of channel to produce impulse response and obtained the postcursor number  $k$  based on channel impulse response or channel characterization. Then, we divided  $M + k$  bits data stream to  $M$  blocks with  $k$  bits in each block and calculated decimal number of every block to obtain the coefficient  $C_{xi}$  in ISI matrix. The PJ matrix is formed based on the data rate and PJ frequency. The PJ frequency is obtained from the jitter sequence spectral analysis. DCD matrix is based on the data transition. Second, according to (10) and (11), the initial estimation jitter component value could be obtained by LS. The final estimation of PJ, ISI, DCD, and RJ values is based on (12)–(14), respectively.

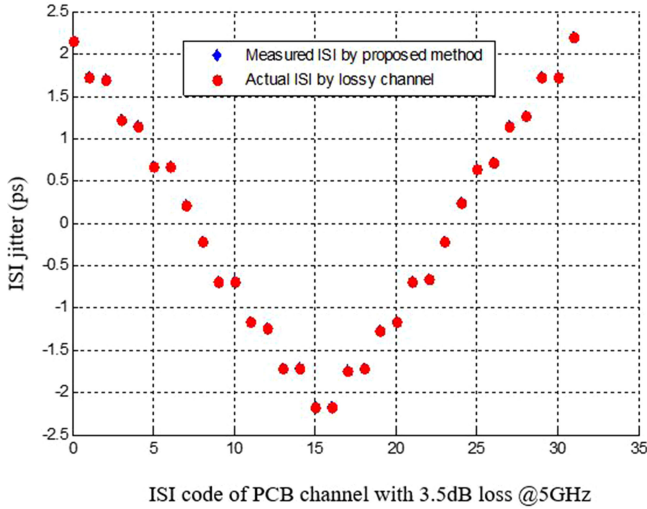


Fig. 7. Comparison of ISI TIE binary combination and estimation result at 10 Gb/s.

#### IV. SIMULATION

In this section, we used MATLAB simulation to validate the proposed new ISI modeling and decomposition method. The PRBS7 data length is 1.27 kb and the data rate is 10 Gb/s in the simulation and can be extended to 25 Gb/s. The PJ was a sine wave with 100 MHz frequency at 10 Gb/s and 250 MHz frequency at 25 Gb/s, respectively. Channel A with 3 dB and channel B with 5 dB loss at 5 GHz are modeled to verify the ISI model and estimation.

##### A. Validation of the ISI Jitter Estimation

First, we verify that the ISI jitter estimation model proposed by this paper is previously  $k$ -bit dependent and use ISI TIE jitter sequence as reference criteria. We used the extraction of  $S$ -parameter of a print circuit board (PCB) transmission line with insertion loss 3.5 dB at 5 GHz and 7.5 dB at 12.5 GHz to generate two groups of ISI TIE sequences. The postcursor number  $k$  of the transmission line is 5. Each group of ISI jitter sequences is classified to  $2^5$  binary combinations, which is from 00000 to 11111. The corresponding jitter amounts are denoted from  $J_0$  to  $J_{31}$ . The red dots in Figs. 7 and 8(a) represent the TIE ISI jitter in each ISI binary combination from  $J_0$  to  $J_{31}$  at 10 and 25 Gb/s, respectively. They show that different binary combinations correspond to different ISI jitter. The peak-peak ISI is about 4.6 ps at 10 Gb/s and 7.2 ps at 25 Gb/s in this PCB transmission line.

Second, we verify the decomposition algorithm, which can accurately estimate every binary combination from  $J_0$  to  $J_{31}$ . The whole PRBS7 data stream with only ISI jitter was sent to the proposed method. The estimated ISI jitter for each binary combination is represented by blue dots in Fig. 7 for 10 Gb/s and in Fig. 8 for 25 Gb/s. The estimated ISI (pk-pk) value about 4.6 ps at 10 Gb/s and 7.2 ps at 25 Gb/s. The estimation error at 25 Gb/s is fall in  $(-0.025 \text{ ps } 0.025 \text{ ps})$ , as shown in Fig. 8(b). In order to verify ISI model, we also modeled channel A with 3 dB loss ( $k = 3$ ) and channel B ( $k = 6$ ) with 5 dB loss at 5 GHz

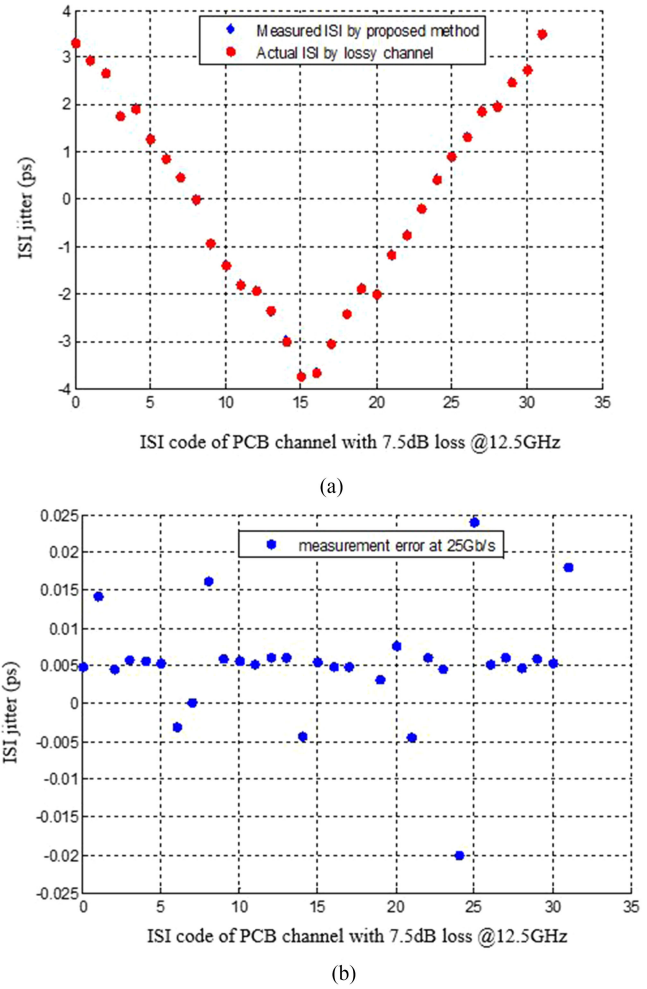


Fig. 8. Comparison of ISI TIE binary combination and estimation result at (a) 25 Gb/s and (b) estimation error.

to verify the ISI model and estimation. Fig. 9(a) and (b) shows the estimation ISI jitter of each binary combination and actual ISI, which demonstrates the accuracy of the ISI estimation for different channels.

##### B. Validation of PJ, DCD, RJ, and ISI Jitter Estimation

In the 10 Gb/s simulation, PRBS7 data stream with different PJ (0 ps, 20 ps, peak-peak value), DCD (0 ps, 4 ps, 8 ps), RJ (0 ps, 2.13 ps, rms value) and ISI jitter caused by channels A and B were sent to the algorithm. The simulation results are summarized in the third column group of Table II. It shows the estimated jitter is very close to the added jitter. The estimation error of ISI is less than 0.6 ps.

In the 25 Gb/s simulation, PRBS-7 data stream with different jitter components was generated. The data stream with different PJ (0, 4 ps peak-peak), DCD (0.2 ps), RJ (0, 1.5 ps) and ISI jitter caused by channel A and PCB transmission line were sent to the algorithm. The simulation results are summarized in the third column group of Table III. The estimation error of ISI is less than 0.5 ps. The estimation error of PJ and RJ is less than 0.3 ps.

TABLE II  
COMPARISON OF THE PROPOSED METHOD AND JNEYE AT 10 GB/s (JITTER UNIT: PS)

| Channel   | Added jitter |      |    |         | Sample data | JNEye (state of the art) |      |      |      | Sample data | The proposed method |      |      |      |
|-----------|--------------|------|----|---------|-------------|--------------------------|------|------|------|-------------|---------------------|------|------|------|
|           | DCD          | RJ   | PJ | ISI     |             | DCD                      | RJ   | PJ   | ISI  |             | DCD                 | RJ   | PJ   | ISI  |
| Channel A | 0            | 0    | 0  | unknown | 8 k         | 0                        | 0    | 0    | 3.5  | 1.27 k      | 0                   | 0    | 0    | 3.6  |
|           | 4            | 2.13 | 20 | unknown | 8 k         | 4.2                      | 2.12 | 20.2 | 3.46 | 1.27 k      | 4.1                 | 2.14 | 20.1 | 3.5  |
|           | 8            | 2.13 | 0  | unknown | 8 k         | 8.2                      | 2.11 | 0    | 3.55 | 1.27 k      | 8.1                 | 2.12 | 20.2 | 3.6  |
| Channel B | 0            | 0    | 0  | unknown | 8 k         | 0                        | 0    | 0    | 9.2  | 1.27 k      | 0                   | 0    | 0    | 9.1  |
|           | 4            | 2.13 | 20 | unknown | 8 k         | 4.1                      | 2.2  | 20.5 | 9.1  | 1.27 k      | 4.1                 | 2.15 | 20.1 | 9.15 |
|           | 8            | 2.13 | 0  | unknown | 8 k         | 8.2                      | 2.16 | 20.3 | 9.15 | 1.27 k      | 8.1                 | 2.12 | 20.2 | 9.1  |

TABLE III  
COMPARISON OF THE PROPOSED METHOD AND JNEYE FOR AT 25 GB/s (JITTER UNIT: PS)

| Channel               | Added jitter |     |    |         | Sample data | JNEye |      |      |      | Sample data | The proposed method |      |      |      |
|-----------------------|--------------|-----|----|---------|-------------|-------|------|------|------|-------------|---------------------|------|------|------|
|                       | DCD          | RJ  | PJ | ISI     |             | DCD   | RJ   | PJ   | ISI  |             | DCD                 | RJ   | PJ   | ISI  |
| Channel A             | 0            | 0   | 0  | unknown | 8 k         | 0     | 0    | 0    | 6.6  | 1.27 k      | 0                   | 0    | 0.07 | 6.5  |
|                       | 2            | 1.5 | 4  | unknown | 8 k         | 2.1   | 1.55 | 4.1  | 6.7  | 1.27 k      | 2                   | 1.54 | 4.01 | 6.6  |
| PCB transmission line | 0            | 0   | 0  | unknown | 8 k         | 0     | 0    | 1.56 | 7.58 | 1.27 k      | 0                   | 0    | 1.63 | 7.61 |
|                       | 2            | 1.5 | 4  | unknown | 8 k         | 2.1   | 1.52 | 4.48 | 7.58 | 1.27 k      | 2.05                | 1.51 | 3.9  | 7.24 |

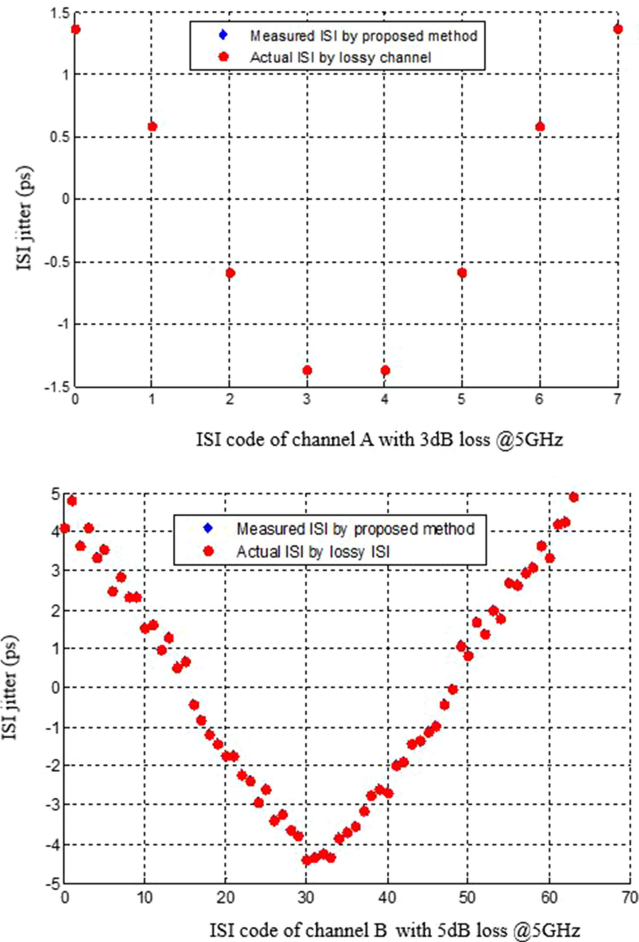


Fig. 9. Comparison of ISI TIE binary combination and estimation results for (a) channel A and (b) channel B at 10 Gb/s.

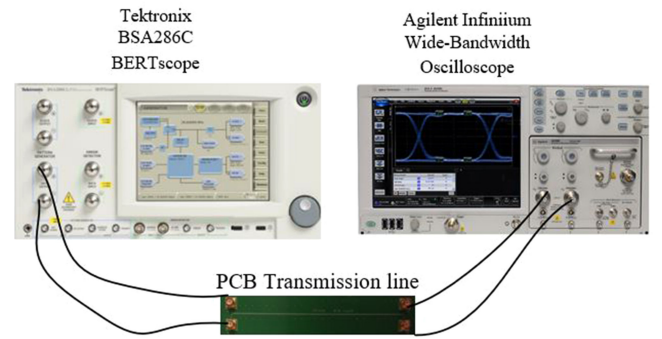


Fig. 10. Experimental test bench.

## V. MEASUREMENT RESULT

To verify the ISI modeling and the accuracy and efficiency of the algorithm, we used a hardware test bench (shown in Fig. 10) to measure the jitter components. A Tektronix BSA286C BERTscope was used to generate data stream with PJ and RJ. An Agilent Infiniium wide-bandwidth Oscilloscope was used to measure the jitter with internal software. The same PCB transmission line in simulation part A was used to generate the ISI jitter. All experiments were done at a data rate of 10 Gb/s.

We also used the result of JNEye with conventional decomposition algorithms as reference to compare the proposed method. JNEye is the state-of-the-art jitter and noise link analysis tool of Intel for evaluate high-speed serial link performance. The jitter decomposition of this platform is based on hybrid algorithms with statistical, frequency-domain, and time-domain analysis with dedicated jitter components modeling and extraction. The accuracy of JNEye has been validated with both simulation and

TABLE IV  
COMPARISON OF THE PROPOSED METHOD /INSTRUMENT/JNEYE FOR PCB TRANSMISSION LINE AT 10 GB/S (JITTER UNIT: PS)

| Added jitter (ps) |      |    |         | Oscilloscope (sample data: 200 k) |      |      |     | JNEye (sample data :8 ) |      |      |      | The proposed method (sample data: 1.27 k) |      |       |      |
|-------------------|------|----|---------|-----------------------------------|------|------|-----|-------------------------|------|------|------|-------------------------------------------|------|-------|------|
| DCD               | RJ   | PJ | ISI     | DCD                               | RJ   | PJ   | ISI | DCD                     | RJ   | PJ   | ISI  | DCD                                       | RJ   | PJ    | ISI  |
| 0                 | 0    | 0  | unknown | 0.8                               | 1.41 | 0.07 | 5.1 | 0                       | 0    | 0    | 4.4  | 0                                         | 0    | 0     | 4.6  |
| 0                 | 0    | 20 | 0       | 0.79                              | 1.51 | 20.5 | 3.2 | 0                       | 0    | 0    | 20.1 | 0                                         | 0.03 | 20.2  | 0    |
| 0                 | 0    | 20 | unknown | 0.8                               | 1.5  | 20.3 | 6.5 | 0                       | 0    | 20   | 4.4  | 0                                         | 0.03 | 20.1  | 4.48 |
| 0                 | 2.13 | 20 | 0       | 0.8                               | 2.9  | 17.6 | 3.2 | 0                       | 2.18 | 20.5 | 0.83 | 0                                         | 2.12 | 20.04 | 0.4  |
| 0                 | 2.13 | 20 | unknown | 0.81                              | 2.49 | 19.8 | 6   | 0                       | 2.18 | 20.5 | 4.6  | 0                                         | 2.12 | 20.04 | 4.59 |

measurement correlations [24]. All tests for comparison were done at 10 Gb/s and 25 Gb/s.

#### A. Validation of the ISI Jitter Modeling

Since the ISI jitter (peak–peak) generated by the PCB transmission line is unknown, we used the oscilloscope to measure the ISI jitter as a reference. In the experiment, PRBS7 without jitter generated by BERTscope was sent to the transmission line and the output of transmission line was connected to oscilloscope to obtain the ISI jitter value.

In the proposed method, the TJ sequences only with ISI jitter were sent to the algorithm and the estimation ISI value was obtained. The comparison is listed in first row of Table IV. In the added jitter term, the unknown for ISI means that the ISI jitter amount is unknown when the transmission line is added. The ISI result in the measurement is 5.1 ps and the one in proposed method is 4.6 ps and JNEye is 4.4 ps. These values are very close. Therefore, the ISI modeling is validated. RJ (rms value) in the simulation is 0 ps while the one in the measurement is 1.41 ps due to the instrument noise. PJ in the proposed method is 0.07 ps (peak–peak) while the oscilloscope result is about 0.07 ps.

#### B. Comparison of the Accuracy and the Sample Data

In the experiment, PRBS7 with different PJ (0 ps, 20 ps, peak–peak), RJ (0 ps, 2.13 ps, rms) generated by BERTscope was sent to the PCB transmission line. The same amount of PJ, RJ, and ISI were added to the TJ sequence in the simulation.

Table IV shows the comparison results among oscilloscope, JNEye, and the proposed method. When the measurement result is stable, the experiment should run several minutes according to test experience. When the added ISI is 0, the oscilloscope result of ISI jitter is about 3.2 ps due to the cable while in the proposed algorithm the estimation is about 0 ps.

The DCD measurement result is 0.8 ps while in the proposed method is 0. The data sample in each simulation is 1.27 kb, but the oscilloscope requires at least 200 k data in the experiment. Therefore, the proposed algorithm has comparable accuracy using fewer data samples than the instrument and have the same accuracy compared to JNEye.

We also modeled channels A and B to verify the proposed method for 10 Gb/s. With different ISI, the comparison of JNEye and the proposed method is listed in of Table II. It shows that both of two methods have the same estimation accuracy

with the same sample data at 10 Gb/s. For the 25 Gb/s comparison, PRBS7 data stream with different PJ (0 ps, 4 ps pk–pk), RJ (0 ps, 1.5 ps), DCD (0 ps, 2 ps) and ISI jitter generated from PCB transmission line and channel A with 6 dB loss at 12.5 GHz were sent to JNEye and the proposed method. Table III shows the estimation results in the proposed method and the JNEye estimation result. Both methods have the same estimation accuracy with different sample data. However, the proposed method has two obvious advantages. First, the sample data is less in proposed method. Second, the proposed method can provide ISI analysis in detail for each ISI binary combination.

## VI. CONCLUSION

We developed an efficient and accurate jitter decomposition algorithm that simultaneously extracts the periodic jitter, random jitter, ISI jitter, and duty cycle distortion in high-speed link. The proposed ISI jitter model applied in this paper is based on channel impulse response in time domain, which is much simpler than the conventional cursor convolution technique applied in most of the state of the art. Also, the new ISI jitter model provides much more detailed ISI jitter information than conventional methods. Different channels simulation results demonstrate the correctness and accuracy of new ISI model at 10 and 25 Gb/s, respectively. The algorithm utilizes fewer sample data than the state of the art while maintaining great estimation accuracy in both clock pattern and data pattern. A great deal of simulation results and hardware measurements demonstrates the accuracy of the proposed jitter measurement method. However, there are still limitations in the proposed method. First, the algorithm needs the TIE sequences as an input which requires extra instruments such as TIA to store the jitter sequence. Second, the proposed method is applied in low lossy channels, which guarantee that the eye diagram should be open before the receiver of the system.

## REFERENCES

- [1] M. P. Li, *Jitter, Noise and Signal Integrity at High-Speed*. Englewood Cliffs, NJ, USA: Prentice-Hall, 2007.
- [2] K.-H. Cheng, J.-C. Liu, C.-Y. Chang, S.-Y. Jiang, and K.-W. Hong, "Built-in jitter measurement circuit with calibration techniques for a 3-GHz clock generator," *IEEE Trans. Very Large Scale Integr. Syst.*, vol. 19, no. 8, pp. 1325–1335, Aug. 2011.
- [3] J.-C. Hsu and C. Su, "BIST for measuring clock jitter of charge-pump phase-locked loops," *IEEE Trans. Instrum. Meas.*, vol. 57, no. 2, pp. 276–285, Feb. 2008.



- [4] M. Li, J. Wilstrup, R. Jessen, and D. Petrich, "A new method for jitter decomposition Through its distribution tail fitting," in *Proc. IEEE Int. Test Conf.*, Sep. 1999, pp. 788–794.
- [5] K. Nose, M. Kajita, and M. Mizuno, "A 1-ps resolution jitter-measurement macro using interpolated jitter oversampling," *IEEE J. Solid-State Circuits*, vol. 41, no. 12, pp. 2911–2920, Dec. 2006.
- [6] G. Hnsel, K. Stieglbauer, G. Schulze, and J. Moreira, "Implementation of an economic jitter compliance test for a multi-gigabit device on ATE," in *Proc. IEEE Int. Test Conf.*, pp. 1303–1312, Oct. 2004.
- [7] D. Hong and K.-T. Cheng, "An accurate jitter estimation technique for efficient high speed I/O Testing," in *Proc. IEEE Asian Test Symp.*, Oct. 2007, pp. 224–229.
- [8] F. Nan, Y. Wang, F. Li, W. Yang, and X. Ma, "A better method than tail-fitting algorithm for jitter separation based on Gaussian mixture model," *J. Electron. Test., Theory Appl.*, vol. 25, no. 6, pp. 337–342, Dec. 2009.
- [9] R. Stephens, "Separation of a random component of jitter and a deterministic component of jitter," U.S. Patent 7 191 080, Mar. 13, 2007.
- [10] C.-K. Ong, D. Hong, K.-T. Cheng, and L.-C. Wang, "Jitter spectral extraction for multi-gigahertz signal," in *Proc. Asia South Pac. Des. Autom. Conf.*, Jan. 2004, pp. 298–303.
- [11] V. K. Sharma, J. N. Tripathi, R. Nagpal, S. Deb, and R. Malik, "A comparative analysis of jitter estimation techniques," in *Proc. IEEE Int. Conf. Electron., Commun., Comput. Eng.*, 2014, pp. 125–130.
- [12] H. Pang, J. Zhu, and W. Huang, "Jitter decomposition by fast Fourier transform and time lag correlation," in *Proc. IEEE Int. Conf. Commun., Circuits, Syst.*, Jul. 2009, pp. 365–368.
- [13] T. Yamaguchi *et al.*, "An FFT-based jitter separation method for high-frequency jitter testing with a 10x reduction in test time," in *Proc. IEEE Int. Test Conf.*, Oct. 2007, pp. 1–8.
- [14] Q. Dou and J. Abraham, "Jitter decomposition by time lag correlation," in *Proc. IEEE Int. Symp. Quality Electron. Des.*, Mar. 2006, pp. 6–10.
- [15] J. Zhu and W. Huang, "Jitter analysis and decomposition based on EMD/HHT in high-speed serial communications," in *Proc. IEEE Int. Conf. Test. Diagnosis*, Apr. 2009, pp. 1–4.
- [16] D. Oh and X. Yuan, *High Speed Signaling: Jitter Modeling Analysis, and Budgeting*. Englewood Cliffs, NJ, USA: Prentice-Hall, 2011, ch. 3, pp. 44–64.
- [17] L. Xu, Y. Duan, D. Chen, "A low cost jitter separation and characterization method," in *Proc. IEEE 33rd VLSI Test Symp.*, May 2015, pp. 1–5.
- [18] Agilent Technol., "Analyzing jitter using Agilent EZJIT plus software," Agilent Technol., Santa Clara, CA, USA, Appl. Note AN-5989-3776EN, 2005.
- [19] Y. Duan and D. J. Chen, "Accurate jitter decomposition in high speed link," in *Proc. IEEE Int. Symp. VLSI Test Symp.*, Apr. 2017, pp. 1–6.
- [20] A. Dey and H. J. Song, "Modeling of ISI in high speed serial I/Os terminated with discontinuities," in *Proc. IEEE Int. Symp. Elect. Perform. Electron. Packag. Syst.*, Dec. 2011, pp. 195–198.
- [21] J. Buckwalter, B. Analui, and A. Hajimiri, "Predicting data dependent jitter," *IEEE Trans. Circuits Syst. II*, vol. 51, no. 9, pp. 453–457 Sep. 2004.
- [22] J. M. Cioffi and T. Kailath, "Fast recursive least squares transversal filters for adaptive filtering," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-32, pp. 304–337, Apr. 1984.
- [23] H. Liu and Z. He, "A sliding-exponential window RLS adaptive algorithm: Properties and applications," *Signal Process.*, vol. 45, pp. 357–68, 1995.
- [24] M. Li, M. Shimanouchi, and H. Wu, "Advancements in high-speed link modeling and simulation," in *Proc. IEEE Custom Integr. Circuits Conf.*, 2013, pp. 1–8.



Jose, CA, USA. Her research interests include analog-to-digital convertor design and jitter analysis in high speed link.

**Yan Duan** (S'13) received the B.S and M.S degrees from Hunan University, Changsha, China, and the Ph.D degree from the Department of Electrical and Computer Engineering, Iowa State University, Ames, IA, USA, in 2017, all in electrical engineering.

From 2006 to 2012, she was Senior Hardware Engineer with the Huawei, Shenzhen, China, a serdes architect intern with the Altera, San Jose, CA, USA, in 2014, and an analog circuit design intern with the Huawei, Santa Clara, CA, USA, in 2015. She is currently an Analog Circuit Engineer with the Intel, San



**Degang Chen** (F'16) received the B.S. degree in instrumentation and automation from Tsinghua University, Beijing, China, in 1984, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of California, Santa Barbara, Santa Barbara, CA, USA, in 1988 and 1992, respectively.

From 1984 to 1986, he was with the Beijing Institute of Control Engineering, a space industry R/D institute. From March to August 1992, he was the John R. Pierce Instructor of Electrical Engineering with the California Institute of Technology. After that, he joined the Iowa State University, Ames, IA, USA, where he is currently an Associate Professor. He was with the Boeing Company in summer 1999 and was with Dallas Semiconductor-Maxim in summer 2001. His research experience includes particulate contamination in microelectronic processing systems, vacuum robotics in microelectronics, adaptive and nonlinear control of electromechanical systems, and dynamics and control of atomic force microscopes. His current teaching and research interests include analog and mixed-signal VLSI integrated circuit design and testing. In particular, he is interested in low-cost high-accuracy test and built-in-self-test of analog and mixed-signal and RF circuits, and in self-calibration and adaptive reconfiguration/repair strategies for enhancing the performance and yield of such circuits.

Dr. Chen was the recipient of the Best Paper Award at the 1990 IEEE Conference on Decision and Control and the Best Transaction Paper Award from the ASME Journal of Dynamic Systems, Measurement, and Control in 1995. He became an A.D. Welliver Faculty Fellow with the Boeing Company in 1999.